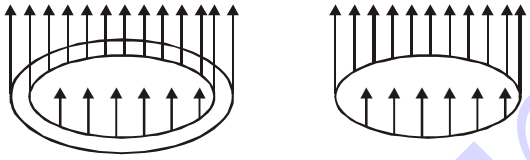
SOLUTION
SECTION-I

Single Correct Answer Type **7 Q. [3 M (–1)]**

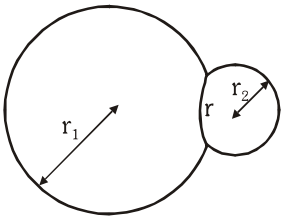
1. Ans. (C)

Sol. Ring have double surface than that of disk.



2. Ans. (B)

Sol.



Let P_0 = atmospheric pressure,

P_1 and P_2 = pressures inside the two bubbles.

$P_1 - P_0 = \frac{4S}{r_1}$, $P_2 - P_0 = \frac{4S}{r_2}$ or $P_2 - P_1 = \frac{4S}{r_2} - \frac{4S}{r_1}$

= pressure difference across the common surface.

Let r = radius of curvature of the common surface.

$\therefore P_2 - P_1 = \frac{4S}{r}$ $\quad \therefore \frac{4S}{r} = \frac{4S}{r_2} - \frac{4S}{r_1}$

3. Ans. (A)

4. Ans. (A)

5. Ans. (B)

6. Ans. (C)

Sol. $R^3 = Nr^3 \Rightarrow r = \frac{R}{N^{1/3}}$

$\Delta U = 4\pi T(R^2 - Nr^2)$

Suppose all this energy is released at the cost of lowering the temperature. If s is the specific heat then the change in temperature would be,

$$\Delta\theta = \frac{\Delta U}{ms} = \frac{4\pi T\left(R^2 - Nr^2\right)}{\frac{4}{3}\pi R^3 \rho s}$$, where ρ is the density,

$$\therefore \Delta\theta = \frac{3T}{\rho s}\left(\frac{1}{R} - \frac{r^2}{R^3}N\right) = \frac{3T}{\rho s}\left(\frac{1}{R} - \frac{r^2R^3}{R^3r^3}\right) = \frac{3T}{\rho s}\left(\frac{1}{R} - \frac{1}{r}\right)$$

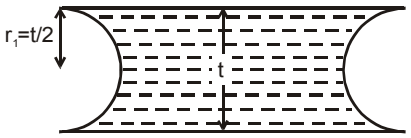
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7. Ans. (A)

Sol. Pressure inside the film is less than outside by an amount, $P = T\left[\frac{1}{r_1}+\frac{1}{r_2}\right]$, where r_1 and r_2 are the radii of curvature of the meniscus. Here $r_1 = t/2$ and $r_2 = \infty$, then the force required to separate the two glass-plates, between which a liquid film is enclosed (figure) is,

$$F = P \times A = \frac{2AT}{t},$$

where t is the thickness of the film, A = area of film.



$$F = \frac{2A^2T}{At} = \frac{2A^2T}{V} =$$

$$\frac{2\times(40\times10^{-4})^2\times(70\times10^{-3})}{0.05\times10^{-6}} = 45 \text{ N}$$

Multiple Correct Answer Type — 6 Q. [4 M (–1)]

8. Ans. (A,C)

Sol. $E=T\,4\pi r^2 \Rightarrow \frac{4}{3}\pi R^3 = n\times\frac{4\pi}{3}r^3 \Rightarrow n=\frac{R^3}{r^3} \Rightarrow R = n^{1/3}r$

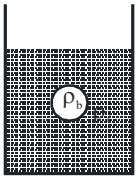
Surface energy of big drop $E'=T4\pi R^2 = T4\pi n^{2/3}r^2 = En^{2/3}$

Energy released = $nE - E' = nE - n^{2/3}E = E(n-n^{2/3})$

9. Ans. (A, D)

Sol. Consider a body of density ρ_b kept in density ρ_ℓ whose viscosity is η and terminal velocity V. Then

$$\vec{F}_{viscous}+\vec{F}_{mg}+\vec{F}_{Buoyancy}=0$$



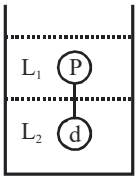
$$\vec{F}_{viscous}+\rho_b\frac{4}{3}\pi R^3(-\hat{j})+\rho\ell\frac{4}{3}\pi R^3(\hat{j})=0$$

$$\therefore\ \vec{F}_{viscous}=(\rho_b-\rho_\ell)\frac{4}{3}\pi R^3(\hat{j}) \Rightarrow 6\pi\eta RV=(\rho_b-\rho_\ell)\frac{4}{3}\pi R^3$$

$\therefore$ if $\rho_b > \rho_l$ then $\vec{F}_{viscous} \uparrow\ V \propto \frac{1}{\eta}$

& if $\rho_b < \rho_\ell$

$\vec{F}_{viscous} \downarrow$



as per given diagram we can say

$\sigma_2 > \sigma_1$; $\rho_1 < \sigma_1$ & $\rho_2 > \sigma_2$

$\Rightarrow \rho_2 > \sigma_2 > \sigma_1 > \rho_1$

$\therefore$ if we put P in L_2 where $|\vec{V}_P| \propto \frac{1}{\eta_2}$ when $\rho_1 < \sigma_2$ $\therefore\ \vec{F}_{viscous} \downarrow$ $\therefore\ \vec{V}_P \uparrow$

$\therefore$ if we put Q in L_1 where $|\vec{V}_Q| \propto \frac{1}{\eta_1}$ when $\rho_2 > \sigma_1$ $\therefore\ \vec{F}_{viscous} \uparrow$ $\therefore\ \vec{V}_P \downarrow$

$$\Rightarrow \frac{|\vec{V}_P|}{|\vec{V}_Q|}=\frac{\eta_1}{\eta_2} \ \&\ \vec{V}_P.\vec{V}_Q<0$$

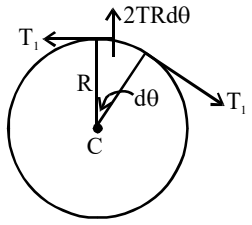
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10. Ans. (B, D)

Sol. (i) due to equal magnitude of surface tension force from all sides, the thread will take circular shape.

(ii)

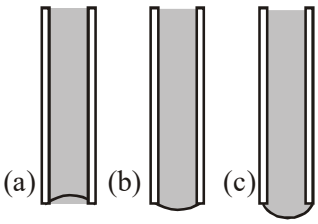


$$2T_1\frac{d\theta}{2}=2TRd\theta$$

$$\Rightarrow T_1=\ 2TR=\frac{2T\ell}{2\pi}=\frac{T\ell}{\pi}$$

11. Ans. (A,B,D)

Sol.



In case (a), it is clear tht the water cannot flow out of the tube. If it could, a perpetuum mobile (Perpetual motion machine) could be established using a paddle rotate ad infinitum by the outflowing water. Cases (b) and (c) are not to simple. The ends of both tubes are lower than the water level and the water pressures there are consequently lower than the atmospheric pressure. In each case, the water wells to such an extent that the between it and the air.

The water surfaces corresponding to cases (a), (b) and (c) are shown in the figure. The greatest curvature

12. Ans. (A,B,C)

Sol. Acceleration will be less than $g\left(\frac{\rho-\sigma}{\rho}\right)$

Time will depend in density ρ. Viscous force may be maximum or minimum both are possible

13. Ans. (B,C)

Sol. $$\frac{F}{2\pi x\ell}=\eta\frac{dv}{dx}$$

$$\frac{F}{2\pi\ell}\int_R^{2R}\frac{dx}{x}=\eta\int_{-v_0}^{2v_0}dv$$

$$\frac{F}{2\pi\ell}\times\ell n2=\eta\times 3v_0$$

$$\frac{F}{\ell}=\frac{6\pi v_0\eta}{\ell n2}$$

$$\frac{F}{2\pi\ell}\times\ell n\left(\frac{x}{R}\right)=\eta v_0$$

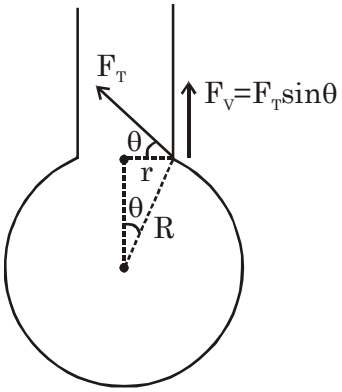
$$\Rightarrow\frac{3}{\ell n2}\ell n\left(\frac{x}{R}\right)=1\ \Rightarrow\ell n\left(\frac{x}{R}\right)=\ell n2^{1/3}\Rightarrow x=2^{1/3}R$$

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Linked Comprehension Type (1 Para × 3Q.) [3 M (-1)]
(Single Correct Answer Type)

14. Ans. (C)

Sol.



$F_T = T\cdot 2\pi r$

$F_V = F_T\sin\theta$

$= T\cdot 2\pi r\cdot\frac{r}{R}$

$= \frac{2\pi r^2 T}{R}$

15. Ans. (A)

Sol. Drop detaches from dropper

When :

$F_V = mg$

$\frac{2\pi r^2 T}{R} = \rho\cdot\frac{4}{3}\pi R^3 g \ ; \ R = \left(\frac{3r^2T}{2g\rho}\right)^{1/4}$

$= \left[\frac{3\times\left(5\times10^{-4}\right)^2\times(0.11)}{2\times10\times1000}\right]^{1/4}$

$\approx 1.4\times10^{-3}$ m

16. Ans. (B)

Sol. Surface energy of drop after detachment = T × surface area

$= T\times 4\pi R^2 = 0.11\times4\times\pi\times(1.4\times10^{-3})^2$

$\approx 2.7\times10^{-6}$ Joule

SECTION-II

Numerical Answer Type Question 1Q.[3M (0)]
(upto second decimal place)

17. Ans. 80.72

Sol. $P_1 - \frac{2T}{r} + \rho gh = P_0$

$h = \frac{P_0 - P_1 + \frac{2T}{r}}{\rho g}$

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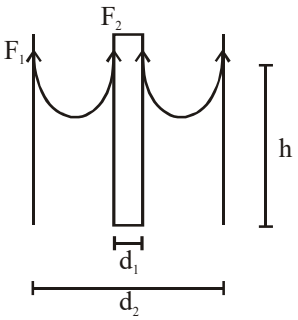
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SECTION-III

Numerical Grid Type (Ranging from 0 to 9) **5 Q. [4 M (0)]**

18. Ans. 6

Sol.



$F_1 = T(\pi) d_2$

$F_2 = T(\pi) d_1$

$F_1 + F_2 = mg$

$$T\pi (d_1 + d_2) = \rho h \frac{\pi}{4}\left(d_2^2 - d_1^2\right) g$$

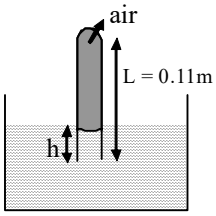
$$h = \frac{4T}{\rho g\left(d_2 - d_1\right)}$$

19. Ans. 1

Sol. If final pressure of gas in tube is P_2 then

$$P_2 - \frac{2T}{r} = P_0$$

(as levels inside and outside are same)



i.e. $P_2 = P_0 + \frac{2T}{r}$

but as temperature remains constant

$\therefore \quad P_1V_1 = P_2V_2$

$$P_0AL = \left(P_0 + \frac{2T}{r}\right) A(L - h)$$

On solving, $h = 0.01$ m

20. Ans. 4

Sol. $$p = \frac{2s}{r} = \frac{BdV}{V}$$

$$V = \frac{4}{3}\pi r^3$$

$$dV = 4\pi r^2 dr$$

$$\frac{2s}{r} = B \times \frac{4\pi r^2 dr}{4/3\pi r^3}$$

$$dr = \frac{2s}{3B} = \frac{2\times 0.075}{3\times 1.25\times 10^8} = \frac{0.15}{3\times 1.25} \times 10^{-8} = 4\text{Å}$$

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21. Ans. 9

Sol. $2\pi r\ell s \geq 4\pi R^2 s$

$r\ell \geq 2R^2$

$$\frac{4}{3}\pi R^3 = \pi r^2 \ell$$

$$R^2 \leq \sqrt{\frac{r\ell}{2}} \; ; \; \frac{4}{3}\left(\frac{r\ell}{2}\right)^{3/2} \leq r^2\ell$$

$$\frac{4}{3}\times\frac{1}{2\sqrt{2}}\times r\sqrt{r}\times\ell\sqrt{\ell} \leq r^2\ell \; ; \; \ell \leq \frac{9r}{2}$$

22. Ans. 2

Sol. $F = \frac{\eta A v}{d}; d = \frac{\eta A v}{F} = \frac{0.01\times10^{-1}\times1^2\times4}{0.002} = 2m$

SECTION-IV

Matrix Match Type (4 × 5) **1 Q. [8 M (for each entry +2(0))]**

23. Ans. (A) P, R,S, T ; (B) P, Q, T ; (C) R, S (D) Q

Sol. (A)$\theta < \theta_R$

Block will not slide

Block in equilibrium

Force exerted by A = Mg

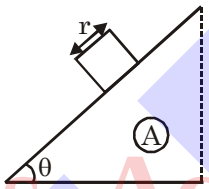
$N = mg\cos\theta$

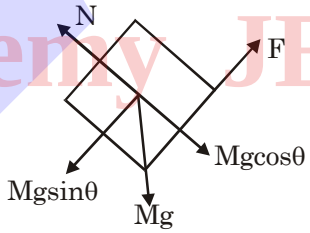
$f = mg\sin\theta$

$F_A = \sqrt{N^2 + f^2} = mg$

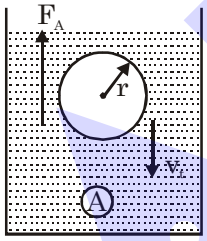
work done by $F_A = 0$ (displacement is zero)

Hence, mechanical energy as well as KE remains const.





(B)



Body falling with terminal velocity

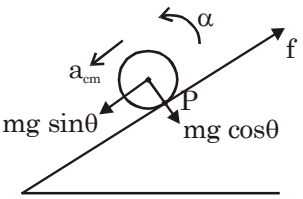
KE is const.

Hence, $F_{net} = 0$

$F_A = mg$

If density of ball increased then mass will increase and it will start accelrating since it acquire new terminal velocity.

(C) mg sinθ



$mg\sin\theta - f = ma_{cm}$ .....(i)

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$f(R) = \left(\frac{2}{5}MR^2\right)\alpha$ .....(ii)

$\alpha = \frac{a_{cm}}{R}$ .....(iii)

On solving

$f = \frac{2}{7}mg\sin\theta$

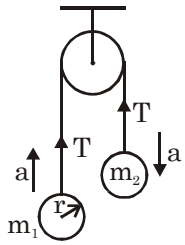
$F_A = \sqrt{f^2 + N^2} = \sqrt{mg\cos\theta^2 + \left(\frac{2}{7}mg\sin\theta\right)^2} < mg$

velocity of pt. P = 0

Hence, $\vec{s}_P = 0$

work done by $F_A = 0$.

(D)



$m_1 < m_2$

$m_2g - T = m_2a$

$T - m_1g = m_1a$

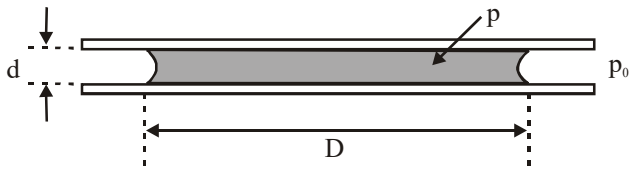
$a = \frac{(m_2 - m_1)g}{m_1 + m_2}$; $T = \left(\frac{2m_1m_2}{m_1 + m_2}\right)g \neq mg$

If density of body is changed, then mass will change and so will acceleration.

Subjective Type **5 Q. [4 M (0)]**

24. **Ans.** $\frac{\pi D^2 Ys}{2d}$

Sol. The cross-sectional edge of the disc of water is a semicircle of radius $r = \frac{1}{2}d$ (see figure). Thus, the curvature of the surface of the water is 2/d, which corresponds to a pressure of curvature of $\Delta p = 2\gamma/d$, where Y is the surface tension. (The other component of the curvature is negligible because D >> d.)



The pressure inside the disc is therefore $p_0 - 2\gamma/d$ when the atmospheric pressure is p_0. This pressure difference acts over a surface area between the water and each of the glass surfaces of $\pi D^2/4$. This implies that a force,

$F = \frac{\pi D^2}{4}\frac{2\gamma}{d}$

'pulls' the glass plates together.

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Note : If d is much smaller than D, this force can be quite considerable. It is in fact very difficult (if not impossible) to separate two parallel glass plates by pulling them in a direction perpendicular to their common plane when there is water between them. In order to be separated, they have to be slid in a direction parallel to that plane.

25. **Ans.** $\left(\frac{R}{24d}\right)^{\frac{1}{3}}$

Sol. $P_{excess}/bubble = \frac{4T}{R}$

$\rho(4\pi R^2)d = \rho\left(\frac{4\pi}{3}r^3\right)$

$r = [3R^2d]^{1/3}$

$P_{excess}/drop = \frac{2T}{r}$

$$\frac{P_{excess}/drop}{P_{excess}/bubble} = \frac{\frac{2T}{r}}{\frac{4T}{R}} = \frac{R}{2r} = \frac{R}{2\left[3R^2d\right]^{1/3}} = \left[\frac{R}{24d}\right]^{1/3}$$

26. **Ans.** $\frac{\lambda a}{2y}g$

Sol. By force balance on the wire,
$2T\cos\theta\ell = \lambda\ell g$

$$\frac{2Ty}{\sqrt{a^2+y^2}}\ell = \lambda\ell g$$

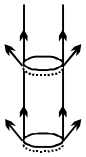
$$\left[T = \frac{\lambda ag}{2y}\right]$$

27. **Ans.** $\mathbf{h = \frac{4T}{r\rho g}}$

Sol. $2[T\cos\theta) \times 2\pi r] = (\pi r^2)\,h\rho g$

$$\frac{4T\cos\theta}{r\rho g} = h$$

$\theta = 0 \longrightarrow$ for wetting completely



$$\frac{4T}{r\rho g} = h$$

28. **Ans. 6**

Sol. $\sigma = \mu\frac{\partial u}{\partial y}$

$= 1.5 \times 4 - 2y$

$= 1.5 \times 4 = 6$

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